

Backward Design

*Practical, assessment-driven course design based on **Understanding by Design** by Wiggins & McTighe.*

CMSE Curriculum Development Sprint

Devin Silvia

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Session roadmap

1. Why backward design?
2. The three-stage *Understanding by Design* (UbD) logic
3. What “assessment-driven design” adds
4. Common traps and design moves
5. A worked example
6. What to try in your own teaching/curriculum development

Note: parts of this content is borrowed from the Institute for Scientist and Engineer Educators as part of their Professional Development Program.

Why backward design?

Establishing a common foundation

- Many of us have or will inherit syllabi, assignments, labs, and lectures but we may not have inherited a coherent approach to design.
- This can lead to courses that feel busy or disjointed, lacking alignment with learning goals/objectives.
- "Backward design" gives a way to start from what we want students to **understand** and **transfer**, then build toward that intentionally.

UbD can be especially helpful when designing a new course or working from a pre-existing course that could use a clearer purpose.

Opening reflection

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- A major assignment you already know you want to use?
- Department expectations or an inherited syllabus?

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Backward design does not say these sorts of starting points aren't without value, but they should not drive the design.

The big idea

Backward design means planning with the end in mind.

- First, identify the learning that matters most.
- Next, decide what would count as convincing evidence.
- Then, plan teaching and learning experiences that prepare students to produce that evidence.

The three stages of backward design[†]

- **Stage 1:** Identify desired results
 - What should learners know, understand, and be able to do?
- **Stage 2:** Determine acceptable evidence
 - What would show that they actually achieved those results?
- **Stage 3:** Plan learning experiences
 - What will equip them to succeed on that work?

[†]Wiggins, G. P., & McTighe, J. (2005). "What is backward design?" in Understanding by design, ASCD.

A useful caveat

Backward design is **not** write outcomes first and then mechanically fill in the rest.

It is a way of checking for coherence as you design, asking at each step:

- Are these the right learning goals?
- Do the assessments actually match those goals?
- Does the instruction genuinely prepare students for the assessments?

Remember what is *not* backward design

- Starting with the textbook (and the default chapter sequence)
- Starting with a favorite activity
- Starting with the cool gizmo/demo
- Starting with the syllabus/slides from the last person who taught this
- Starting with the syllabus from when you took the course
- Starting with whatever is already in D2L

Those can all be resources. They just should not be the design logic.

Two common pitfalls

UbD warns against two common design problems.

Activity-focused teaching

Students do interesting things, but the intellectual target is unclear.

Coverage-focused teaching

The course moves through material quickly, but students may not make meaning from it.

In one case the course is engaging but fuzzy; in the other it is organized but shallow.

Why even experts fall into these traps

- Expertise can make goals feel obvious when they are not obvious to learners.
- We often confuse “I taught it” with “students can use it.”
- We often reward completion, attendance, or recall when we really care about reasoning, judgment, or transfer. The former are easier to measure, but the latter are more meaningful/valuable.

Unpacking Stage 1

When UbD says “desired results,” it means more than a topic list.

You are clarifying:

- What is worth understanding?
- What ideas should endure beyond the exam?
- What should students be able to do with what they learned?
- What misconceptions are likely to get in the way?

Stage 1 components

Established goals

Program outcomes, standards, accreditation demands, departmental expectations.

Understandings

What students should come to see as meaningful generalizations or principles.

Essential questions

Questions that organize inquiry and keep big ideas alive.

Knowledge is not enough

UbD distinguishes among:

- **Knowledge:** facts, vocabulary, concepts, procedures.
- **Skills:** what students can perform or apply.
- **Understanding:** making sense of ideas and transferring learning to new situations.

A student can remember a method and still not understand when, why, or whether to use it.

Transfer is the point

If we say students “understand,” we usually mean they can do more than repeat.

They can:

- interpret,
- explain,
- justify,
- apply in a new context,
- recognize limits,
- revise when conditions change.

Essential questions

Good essential questions do not just decorate a syllabus.

They help students wrestle with issues like:

- What counts as evidence?
- When is a model useful, and when does it break down?
- How do experts decide among competing explanations?
- What makes a solution elegant, ethical, or credible?

Unpacking Stage 2

Before planning activities, ask:

- What would convince me that students reached the goal?
- What evidence would be thin, partial, or misleading?
- What misunderstandings might produce a superficially correct answer?

This is where backward design becomes more demanding than a content outline.

Thinking about evidence

Stage 2 is not just “write the test.”

It usually includes multiple forms of evidence:

- performance tasks,
- exams or quizzes,
- written explanations,
- discussions,
- drafts and revisions,
- reflections or self-assessments.

A practical evidence test

Ask of any assignment, exam item, or discussion prompt:

1. Does this elicit the thinking I care about?
2. Could students succeed through memorization or pattern-matching alone?
3. Would weak understanding still look successful here?
4. Will I be able to tell the difference between emerging, solid, and sophisticated performance?

Assessment-driven design

ISEE offers a slightly different approach that keeps UbD's three-stage structure but makes one move more explicit:

- **Stage 1**: define learning outcomes,
- **Stage 2a**: define evidence of understanding,
- **Stage 2b**: design a prompt that elicits that evidence,
- **Stage 3**: design activity components to align.

That extra move (**Stage 2b**) is powerful because it forces us to ask not only **what** students should know, but also **how they will show it**.

Where this helps most

Assessment-driven design is especially helpful when:

- the desired understanding is complex or abstract,
- you want students to engage in inquiry or project-based work,
- students need to explain reasoning, not just provide answers,
- you suspect your current prompts are not actually eliciting the evidence you want.

Why UbD can sometimes fail

A common design mistake is this:

- We choose a worthwhile goal.
- We imagine rich evidence.
- Then we give students a task that does not actually invite that evidence.

If the prompt is weak, the assessment will underperform even when the learning goal is strong.

Example of the structure in context

We can use ISEE's "Light & Shadow" inquiry as an example for thinking about this framework:

UbD stage	Assessment-driven move	Example move
Identify desired results	Define learning outcomes	Explain light/shadow phenomena using the ray model
Determine acceptable evidence	Define evidence of understanding	Break understanding into dimensions; clarify what performance looks like (i.e., make a rubric!)

Why the middle is hard

Defining evidence can be really hard because instructors must unpack:

- what counts as success,
- what dimensions matter,
- what novice errors are predictable,
- what partial understanding looks like.

A misconception to avoid

Assessment-driven design is not a neat one-way line.

The false hope

Outcomes → evidence → prompt → activity, once and done.

The brutal reality (kidding!)

Iterative movement among goals, evidence, prompts, and learning experiences.

Iteration is a feature

As you design, each piece should revise the others.

- A prompt may reveal that the outcome is vague.
- A rubric may reveal that the prompt misses an important dimension.
- An activity may reveal missing prerequisite knowledge.
- Student work may reveal a misconception you never anticipated.

What counts as alignment?

Alignment is stronger when:

- the learning outcome names meaningful performance,
- the evidence directly represents that performance,
- the prompt actually calls for the evidence,
- the learning activities prepare students for that prompt.

Misalignment patterns

If this happens...	It often means...
Students are surprised by the assignment	Stage 3 did not prepare them for Stage 2
The rubric feels disconnected from the prompt	The prompt does not elicit the intended evidence
Students can pass but cannot explain their reasoning	The evidence is too narrow or low-level
Class time feels busy but unfocused	Stage 1 was not clear enough

An example within a familiar context

Imagine a graduate seminar outcome:

Students will be able to evaluate methodological choices in published research and justify an appropriate design for a related study.

That is already more useful than:

Students will understand research methods.

Unpacking the example

For that research-methods outcome, evidence might include:

- a written critique of a published article,
- an annotated methods comparison,
- a proposal memo defending a chosen design,
- an oral explanation responding to tradeoffs and limitations.

Notice that “evidence” here is not just an exam. It is a set of performances that reveal judgment.

Prompting for the evidence

A weak prompt:

- "Summarize the assigned article."

A stronger prompt:

- "Evaluate the authors' methodological choices, identify one key limitation, and justify a revised design choice for a related study question."

The stronger prompt makes the target thinking visible.

Then plan instruction backward

If that is the prompt, students probably need:

- models of strong critique,
- criteria for methodological quality,
- chances to compare designs,
- low-stakes practice with feedback,
- a language for discussing tradeoffs and validity.

What this means in practice

A backward-designed unit usually includes:

- a visible statement of purpose,
- models of target performance,
- criteria students can use,
- formative checks before high-stakes performance,
- opportunities for feedback and revision,
- a sequence that builds toward independent performance.

Backward design doesn't not mean abandoning everything you do now!

- It is not anti-content.
- It is not anti-lecture.
- It is not anti-test.
- It is doesn't prohibit creativity or spontaneity.

It simply asks whether content, teaching methods, and assessments are aligned with desired results.

Why people resist it

Common concerns:

- "This takes too much time."
- "I do not control my course goals."
- "My field requires too much content for this."
- "Students need basics before more open-ended inquiry."
- "I already have working materials."

And, to be fair, backward design does take time up front, but it often saves time later by reducing drift, redundancy, and unclear assessment.

Start simple!

You don't need to overhaul an entire course to get started with backward design.

Start with one:

- module,
- unit,
- assignment sequence,
- lab,
- discussion week,
- assessment.

A possible redesign workflow

1. Pick one important outcome.
2. Rewrite it so the performance is visible.
3. Identify what strong evidence would look like.
4. Check whether your current prompt elicits that evidence.
5. Revise the prompt or rubric.
6. Revise class activities to prepare students for it.

Questions worth asking when thinking about redesign:

- What do I want students to still remember and use a year from now?
- Where do students most often produce shallow but superficially correct work?
- What assignment in my course most needs a clearer evidentiary target?
- Where am I still mistaking activity, coverage, or compliance for learning?

Closing reflection

What is one assignment or module that would improve immediately if you clarified:

- the intended understanding,
- the evidence you really want,
- and the prompt students need in order to show it?

Questions?

Slides available at: <https://msuceri.org/cmse-sprint-2026>